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IoT and Cloud-based Automated Pet Care System

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***Abstract* - Automatic pet feeder system is designed which is used to take care of pets such as cat and dog. The pet feeder system can deliver food, water and monitor the motion of the pet. This machine is equipped with a different embedded components which is helpful to feed food and dispense water without any human intervention. Unfortunately, due to their hectic schedules and limited time at work, they have neglected their pets and have left them hungry. This project mainly designed for people for saving their time and energy by feeding their pets on time and monitoring through the designated application. The monitoring has been done because of internet connection has been provided through the gadget so that the user can observe the pet's feeding on the corresponding Thing speak cloud. The Arduino UNO and the server motor are among the machine's novel components, which are employed in a bottle that may endure for a week with electrical connections. The purpose of this is to feed food for pets automatically based on amount of food available and also on time. Consequently, most of this pet feeders on the market are operated manually, and these feeders not even use IoT methods. Users may use this Pet Feeder automatically, eliminating the need to worry about their dogs. Those who love pets will be loving to select this type of machine for their pets at home. By using this type of machine consumer will also be less concerned about leaving their pets for small period, such as when returning to their hometown or working full-time. Finally, especially in the mechanical business, it is an excellent technique to increase and employ top and local users. This is quite helpful for the people who has pet, and it uses latest technology where it encourages Internet of things along with Embedded system as it can be applicable for local industrial petting upgrading.**

Keywords–Automatic Pet Feeder, Arduino UNO, Servo motor, Load cell sensor, PIR sensor, Wi-Fi module, IoT.

I.INTRODUCTION

IoT stands for Internet of Things which provides connection among various things such as cars, games, buildings, instruments etc. These physical things use some electronic components which can be embedded system hardware parts so that it helps to connect these devices and helps to share the data on the internet. The internet of things allows users or consumers to control the devices or gadgets from the distance which uses their network configurations, and this helps us to various consumers to use this IoT as it gives good accuracy and efficiency.

Current automatic pet feeders on the market satisfy some of our design goals, but they are extremely expensive. The Pet want Smart Feeder by GemTune has a smartphone app that allows for scheduled feeding and system setup. When the pet is nearby and it is time to feed, the Automatic Pet Feeder from Wireless Whiskers uses RFID to release food for the pet. The PetnetSmart feeder is an automated pet feeder with a smartphone app as the main user interface that focuses on delivering better meal amounts for the pet. The smartphone app can change a pet's feeding schedules and track their general eating patterns; however, it can only track one pet at a time and does not handle numerous pets.

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The present research involves a three-part Internet of Things-based pet care system: a food feeder, a water dispenser, and a litter box. Our project is an Internet of Things-based pet monitoring and feeding system. The importance of this title is that it is intended to provide a starting answer to an issue that practically everyone faces. It's tough to avoid human meddling when it comes to pet care while they're busy. As a result, our system will be capable of overcoming the obstacles that humans confront when caring for pets. This Pet Care System is a full system for keeping track of all of your pet's activities while still allowing them to be free. In addition, the project is separated into various modules, each with its own IR function. Pet monitoring door, pet food feeder, and pet collar system are the three.

Physical monitoring of living or nonliving things is now feasible with a mobile application which acts a interface between the user and the that thing. Through this mobile application by clicking once the complete data can be observed, thanks to the Internet of Things (IoT). It enabled devices to connect and communicate with each other via the Internet with very little human intervention. Various systems have been used for both living and non-living topics, and the Internet of Things (IoT) is constantly growing. Plants, household appliances, and domesticated animals have all been used to create real-time virtual monitoring systems. House dogs are considered as part of the family. Dogs and cats provide humans with a four-legged companion, a never-aging playmate, a never-complaining companion, a guardian who will never abandon someone in the face of danger, a cheerleader who will never let you down, and a long and endless list of other reasons why dogs and cats are a part of human life and existence. These brilliant creatures demonstrate their love for their human families by displaying emotions, loyalty, and calming them. Even if they are hundreds of miles apart, every family member has the right to be looked for at any time. "No Pet Should Be Left Behind," according to the researchers' beliefs.

Animals are a pleasant addition to many households; nevertheless, their requirements can occasionally collide with modern lifestyles. Keeping track of how your animal eats, when it eats, and if it is hungry while you are gone may be difficult. Cats and dogs left to their own devices may often wreak havoc. There is a need for a system that takes the guessing out of caring for one's pet while also helping to reduce the overall stress associated with pet ownership.

The Internet of Things (IoT) is a sort of network that links gadgets to the internet utilizing necessary protocols and sensor equipment to regulate data exchanges and communications for smart recognition, position tracking, and monitoring, among other things. Some of its important qualities are interconnectivity, Things-related services, heterogeneity, flexibility to dynamic changes, scalability, safety, and connection. The IoT architecture is utilized in a range of scenarios, with varying levels of scalability, modularity, and IOT deployment possibilities. In recent years, the number of mobile phone users in the Philippines has increased. According to research, there was a compound annual growth rate of free and premium mobile app downloads worldwide from 2012 to 2017.

II. LITERATURE SURVEY

Wen-Chuan [1] have described a "Remote Pet Feeder Operate System" using the MQTT Protocol, in which the user may control the feeder remotely to toss food at their pet using a smartphone. Some feeders additionally have a camera that lets the owner to monitor their pet at home. These devices, on the other hand, are fixed and cannot move. The photographs are also taken from a fixed vantage point, even though dogs may roam around the house.

In “Animal Tracking and Caring Using RFID” [2] and IoT,it highlights how animals are an integral part of human existence and must be cared for. They must be supplied with a proper, acceptable, and safe habitat, as well as correct dietary care. It'll take a system to track and monitor the creatures from afar. This system should be intelligent enough to notify the pet's owner of current events, be able to provide animals with the necessary amount of food and water, maintain the animal's preferred room temperature, live stream the animal's movement and characteristics, and be able to detect the animal at a maximum range.

In "Automatic Pet Monitoring and Feeding System Using IoT,” [3] demonstrate how to create an automatic pet monitoring and feeding system using the Internet of Things. The purpose of using the term in this manner is to provide a universal answer

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to the challenges that everyone in the globe faces. The system's goal is to eliminate human intervention when it comes to pet care in their busy lives. Only by automating the pet-care procedure is this possible. This Pet Care System is a thorough system for keeping track of all your pet's activities. It also gives the pet a sense of freedom. As a result, the project is divided into various modules, each with its own IR feature. Pet monitoring door, pet food feeder, and pet collar system are the three.

In "Smart dog cage system," describe a smart cage system that is used to care for a dog when the owner has a busy schedule and does not have time. Smart dog cages have aided dog owners in making their lives simpler by reducing their anxiety about their pets when they are gone. Furthermore, the system was created for usage in hospitals, as well as for patient care.
An automatic module for cleaning faeces and urine with image processing, an automatic feeding system, and a canine monitoring system are all included in the Smart cage system. A computer processes all these components. A computer processes all these components. This system takes use of an IOS mobile application that connects to the server and allows you to change a parameter, regulate a cage, and observe a dog through an internet network connection.

In "The Study and Application of the IoT in Pet Systems," [4] try to enhance pet appliances by adding location tracking capabilities and to make it easier for pet owners to grow their dogs. Even though there are several automatic pet monitoring systems available, such as automatic pet doors and pet feeds, these systems are unable to satisfy the expectations of pet owners. Most pet-care systems, for example, are based on infrared detection and recognition, which may be used to recognise pets at the entrance, track their activities, and notify owners when they enter places where they are not permitted. Because infrared detectors are easily impacted by many circumstances, this infrared detector design has certain drawbacks. The detection cannot be identified appropriately. According to previous research by the author, own has created a prototype of a pet system based on the cyber-physical-system concept. As a result, the system incorporates IoT infrastructure and applies the cat-raising case study.

The interaction between electronics and actual objects/humans in the real world is detailed by in "Intelligent pet monitor system using the internet of things." Every day, more people are becoming aware of this issue. The concept is to utilise technology to link pet owners with their pets via any device that is accessible from anywhere. Contrary to popular belief, technology has made life easier for owners and allowed them to communicate with their dogs. As a result, the topic of how to rear pets in a simple manner is the primary worry, and the writers attempt to address this issue. The writers of this study focused on the system design to monitor the pets' activities and manage pet eating time using the technology of the Internet of Things.In "Internet of Things for Human-Pet Interaction," [5] It address reasons including ageing and diminishing birth rates. The fast-paced nature of modern civilization necessitates a hectic lifestyle, which has an influence on interpersonal interactions. People choose to have a pet as a friend, therefore. However, most pet owners have a hectic work schedule and are unable to devote their complete attention to their pets [6-12]

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III. PROPOSED METHODOLOGY

The below figure represents the block diagram of pet care system. The block diagram has two modules which are food feeder and water dispenser, Litter system. The food feeder system is internally connected to the Load cell sensor and servo motor. The servo motor and Load cell sensor has connected to the Arduino uno and ESP 8266 Wi-Fi Module which in turn connected to the IoT. By using IoT, Human who is anywhere can watch whether the pet has been ate the food or not by Thing-speak software.

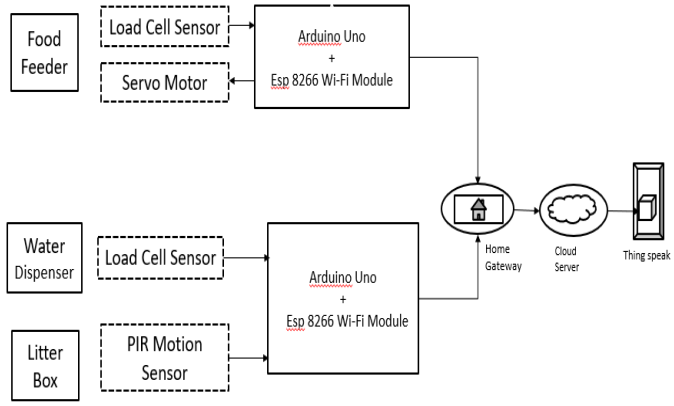

Fig: 3.1 Block Diagram

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The system has used components as:

1. Arduino uno
2. Load Cell Sensor
3. HX711 Amplifier
4. ESP 8266 Wi-Fi Module
5. PIR Motion Sensor
6. LCD display
7. Servo Motor
8. Mobile Phone
9. Push Switch button

Software components:

1. Arduino IDE
2. Things speak cloud

The second module which shows in the block diagram is water dispenser and Litter system. These systems are connected to Load cell Sensor and PIR motion sensor where in turn each of the components are connected by Arduino uno and ESP 8266 Wi-Fi Module. These modules connected to the IoT which is used by Human to see the pet's water dispenser and motion. The Esp 8266 Wi-Fi module provides Internet connection to the microcontroller which it helps us to transfer information using TCP/IP protocol to the Thing Speak Cloud. The user can log-in to Thing Speak app and see the details of the pet's food feeding and water feeding at corresponding time and date.

IV. EXECUTION FLOW

The project consists of two modules: First module represents Food Feeder System, and Second module represents Water Dispenser System and Litter box system. To build sensor reception and transmission, the system takes use of an open-source hardware platform. That design has two advantages: one is that it is simple to maintain, and the other is that it is simple to construct. The system is designed to be scalable, which means it can be applied on a big scale. The food feeder system is internally connected to the Load cell sensor and servo motor. The servo motor and Load cell sensor has connected to the Arduino uno and ESP 8266 Wi-Fi Module which in turn connected to the IoT (Internet of Things). By using IoT, Human who is anywhere can watch whether the pet has been ate the food or not by Thing-speak software.

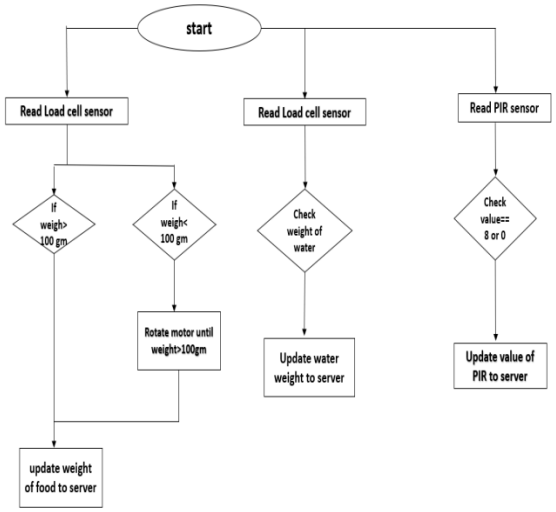

Fig: 4.1 Flow chart

Load cell Sensor is the one which measures weight of food in the plate. If the weight of the food is in between 0 to 100 grams, then the servo motor which is been connected to Arduino will be rotating in the anticlockwise direction with certain speed. From the upper bowl, the food starts dropping into the plate until the weight of the food which is present in the plate will increase up to 200 grams.Once it fills the food then servo motor starts rotating clockwise direction with same speed and closes the mouth of the bowl where leftover food will be safe. The food feeder's software comprises servo rotation, load cell reading and weight computation, network connection, and communication with Thing Speak. First and foremost, we set the servo to its original position. The servo rotates from zero degrees to 180 degrees in one-degree steps based on the weight of the load cell sensor; once it reaches 180 degrees, it rotates 180 degrees backward till it reaches its initial position. If the servo continues in this loop indefinitely.

As a result, each time the button is pressed, the same amount of food is dropped, which we refer to as one portion. Then we'll be able to regulate the amount of food eat.

The second module represents the water dispenser system and litter box system. These systems are connected to Load cell Sensor and PIR motion sensor where in turn each of the components are connected by Arduino uno and ESP 8266 Wi-Fi Module. These modules connected to the IoT which is used by Human to see the pet's water dispenser and motion. In this module the water dispenser system has been connected to load

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cell sensor which is used to measure the weight of the water and updates all the weight of the water to the user present anywhere. The weight of the water is uploaded to the Thing speak cloud. The user can login into the system and can observer whether water has been present or not and he can observe the consumption level of the water also and he make sure that the pet is safe through this system. Using PIR motion sensor it detects the motion of the pet whether it comes or not for in the litter box system. If the pet doesn't come over a day, then the value will be shown as zero otherwise if it comes over a day at the corresponding time it shows value non-zero which I used in the programming as 8 value to be updated.

V. PROTOTYPE AND CLOUD SERVER

The below figure shows the prototype of our project. It shows how
various sensors are connected on the Arduino and in turn connected to IOT server so that all the sensor values can be monitored from anywhere in the cloud using Thing-speak Server. Also, the values can be monitored regularly using the LCD as for prototype.

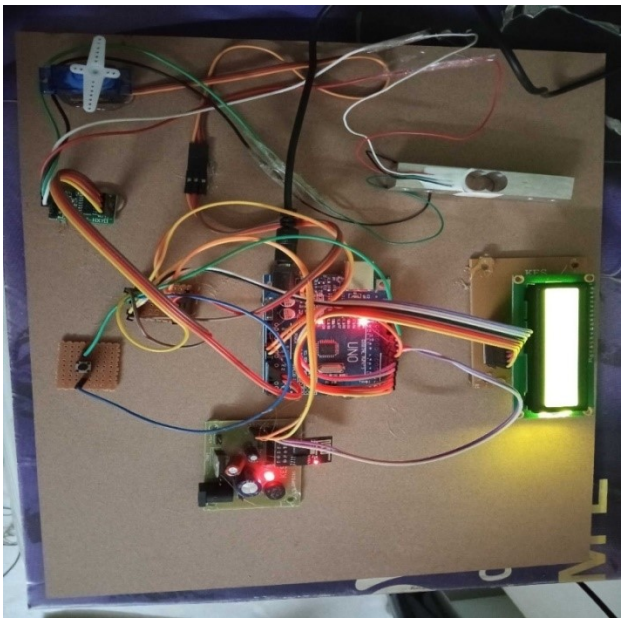

Fig 5.1 Food Feeder System

The food feeder system is internally connected to the Load cell sensor and servo motor. By using IoT, Human who is anywhere can watch whether the pet has been ate the food or not by Thing-speak software. Load cell Sensor is the one which measures weight of food in the plate. If the weight of the food is in between 0 to 100 grams, then the servo motor which is been connected to Arduino will be rotating in the anticlockwise direction with certain speed. From the upper bowl, the food starts dropping into the plate until the weight of the food which is present in the plate will increase up-to 200 grams. Once it fills the food then servo motor starts rotating clockwise direction with same speed and closes the mouth of the bowl where leftover food will be safe.

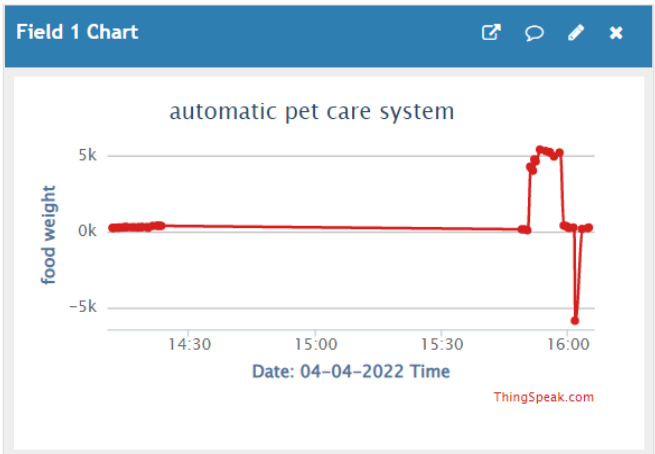

Fig 5.2 Thing Speak Cloud-Food Feeder System

This part displays the output of all systems linked to the internet via an online cloud portal called Thing Speak, which continually monitors the data and updates it every second. The data is constantly checked, and if any changes occur, the programmed, which was designed with a local IP address in mind, monitors and reports on them. Arduino is utilized as a gateway in this situation. An extra application is installed and setup in the android host. The result shows the weight of the food which is present in the plate which is calculated by load cell sensor. The Load Cell Amplifier is a compact breakout board for the HX711 IC that makes reading load cells for weight measurement simple. You can read changes in the resistance of the load cell by connecting the amplifier to your micro-controller, and with simple calibration, you can get preciseweights. In this result the user can observe the values based on the date and time correspondingly. The user can observe the weight of the food in different format. The one which is shown is graphical format. But the supported formats

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are spline, bar, unit step, column etc. Based on user's compatibility he can select his options. In the result the graph is also showing the negative values it is because that the load cell sensor which consists of both positive and negative terminals so when I was measuring the weight, I exchanged the terminals, so it has given value of negative. The load cell sensor measures maximum of five kilograms of weight.

The below prototype is of water dispenser system and litter box system. The working model of water dispenser system is same as that of food dispenser system. So, there is not much change in that but as per litter box system it gives result based on the pet's movement in the litter box system. This module comprises of the water dispenser system has been connected to load cell sensor which is used to measure the weight of the water and updates all the weight of the water to the user present anywhere. The weight of the water is uploaded to the Thing speak cloud.

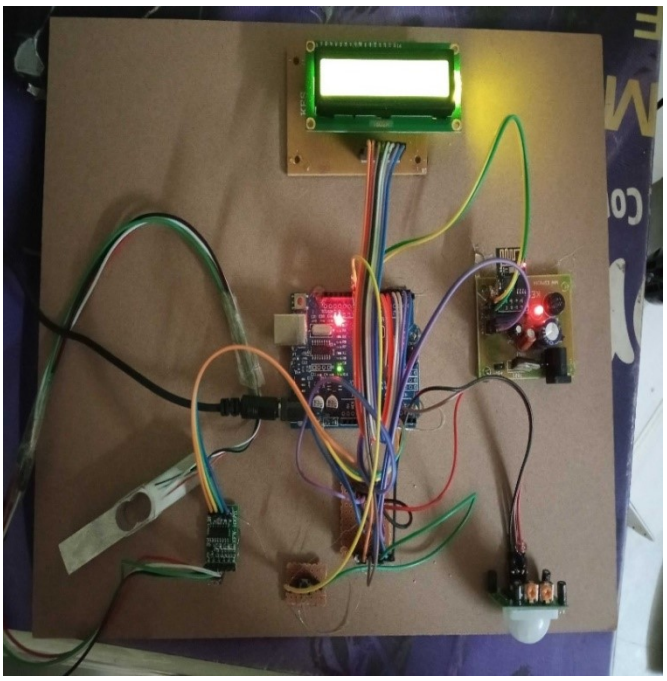

Fig 5.3 Water Dispenser System

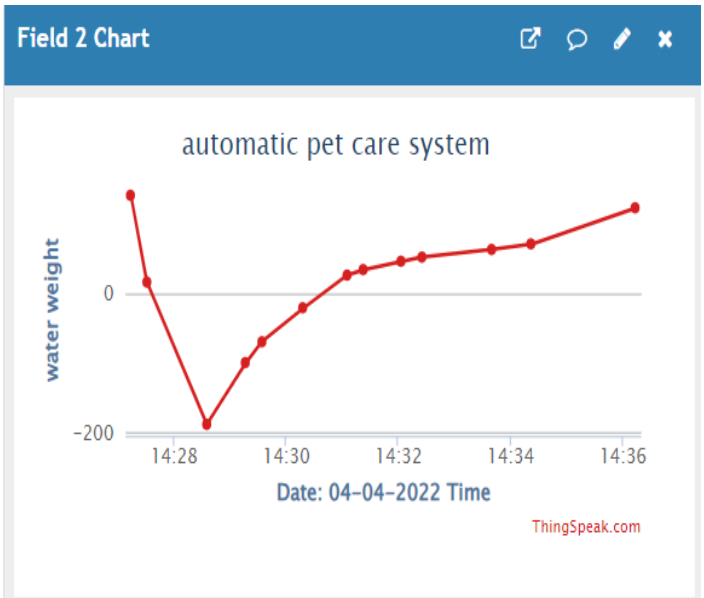

Fig 5.4 Thing speak cloud-Water Dispenser System

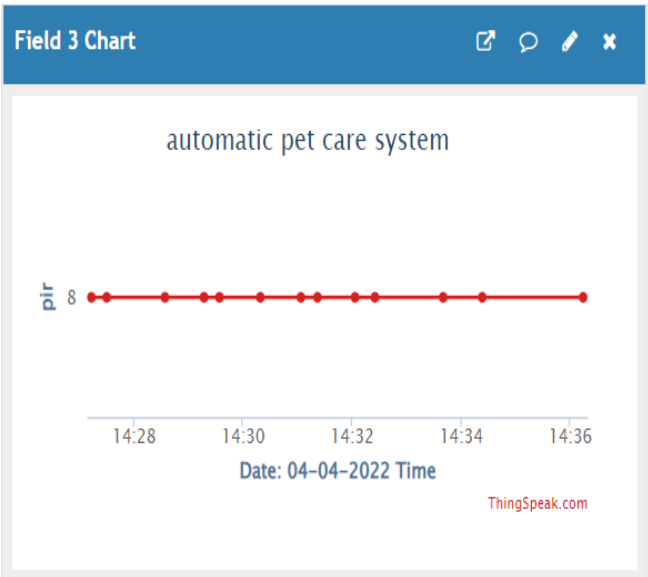

Fig 5.5 Thing speak cloud-PIR motion sensor

VI. CONCLUSION AND FUTURE SCOPE

This research work has successfully developed an IoT-based pet care system by using a variety of sensors and actuators (food feeder, water dispenser, and litter box). Food consumption monitoring and rapid and remote food distribution are among the features of the food feeder subsystem. The water dispenser can keep

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track of how much water is used. The litter box keeps track of how often and when the pet goes to the bathroom. We utilize a smartphone interface to operate and monitor the devices as well as show the statistical information in order to combine the three subsystems.

Based on the current paper, we have recommended certain changes. For starters, because there are multiple established smart feeders on the market, more functions may be added. Food feeders, such as those that feed on a set timetable. Second, if on the food feeder or the water dispenser, any unexpected action happens. Such as food stuck at the exit, or food or water dispenser We need an alert system to inform pets if water spills out. Owners in a timely manner Finally, we may look at the statistics chart. Use big-data technologies and data analytic techniques to Data should be formatted and analysed further. Therefore, any odd behaviour in a pet may be noticed and reported to the owners, and a health analysis report can be prepared on a regular basis. Furthermore, we may improve performance by enhancing the system connection's stability, the device modules' longevity, and the data's correctness. Finally, because there have been numerous previous products and works on pet care from various perspectives, such as smart cages, smart toys, wearable devices, and even smart homes, we can devote more resources to connecting more devices to enrich and enhance the pet care system and meet more demands from pet owners.

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